

Title: Option Momentum: Further Evidence

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Option Strategy Momentum: Further Evidence

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Abstract

In this study, we investigate the presence of momentum effects within three popular delta-neutral option trading strategies: straddles, strangles, and butterfly spreads. Our analysis primarily focuses on options associated with stocks of large-cap and blue-chip firms that constitute the S&P 100 Index over the period spanning 2000 to 2019. We find strong momentum patterns in the returns of straddle and strangle strategies, with the formation period spanning one to three years, excluding the most recent month. However, the butterfly spreads strategy does not demonstrate similar momentum effects. These findings remain robust after accounting for conventional options risk characteristics, and the observed momentum patterns can not be explained by traditional risk factor models. We also find that straddle and strangle strategies yield even stronger momentum returns within the neutral context of these factors.

1 Introduction

Momentum, commonly known as the tendency of assets that have exhibited strong or weak performance in the past to maintain such performance in the future, is a widely studied phenomenon in financial markets. In their seminal paper, Jegadeesh and Titman (1993) document the profitability of trading strategies involving buying past winners and selling past losers in the U.S. stock market. Since then, there has been growing evidence affirming the existence of the momentum phenomenon across diverse markets and asset classes. Within the stock market, momentum effects are observed in industry portfolios (Moskowitz and Grinblatt (1999)), global stock returns (Rouwenhorst (1998), Griffin et al. (2003), Chui et al. (2010), and Fama and French (2012)), and factor returns (Arnott et al. (2021)). Momentum is also identified in the foreign exchange market (Okunev and White (2003), Menkhoff et al. (2012), and Zhang (2022)), corporate bonds (Jostova et al. (2013) and Li and Galvani (2021)), and commodity markets (Miffre and Rallis (2007) and Boons and Prado (2019)).

The momentum phenomenon, extensively studied in the spot markets, has received comparatively less attention in the options market. Early studies have highlighted mispricing within stock index options, indicating the existence of a time series momentum (see, for example, Stein (1989) and Poteshman (2001)). A more recent study by Heston et al. (2023) examines the existence of the cross-sectional momentum effect within the options market. Their findings document significant momentum effects in the returns of delta-neutral straddles on individual U.S. equities. Their analysis, which encompasses a broad spectrum of optionable stocks, brings both advantages and challenges. On one hand, the large sample size, comprising approximately 1,368 individual equities each month, provides comprehensive insights. On the other hand, it raises concerns, particularly regarding the liquidity of options linked to small or medium-sized firms, which form a substantial part of the sample. These options often exhibit limited liquidity compared to those of larger firms, posing challenges in portfolio rebalancing for momentum strategies. In addition, given Banz (1981)'s findings that smaller firms generally yield higher risk-adjusted returns compared to larger firms, the question arises as to whether an analogous size-factor effect exists within the options markets. In other words, it remains unclear whether smaller firms, particularly those with less liquid options, are major contributors to the observed momentum effect. Furthermore, while Heston et al. (2023) documents a robust momentum effect using the straddle options strategy, it remains uncertain whether this momentum effect persists across other prevalent trading strategies such as

strangles and butterfly spreads. This ambiguity highlights the necessity for further investigation to ascertain the scope and applicability of momentum across various options trading strategies.

In this paper, we extend the work of Heston et al. (2023) and fill the aforementioned gaps by examining the robustness of the cross-sectional momentum phenomenon across various option trading strategies across a range of formation periods. Our study specifically focuses on options contracts linked to equities from large, well-established firms. We select constituents from the S&P 100 Index for our analysis, as these firms encompass a diverse spectrum of industries and represent the prominent large-cap blue-chip companies in the U.S. market. Notably, the equities of these firms often yield relatively lower risk-adjusted returns, yet offer enhanced liquidity in their options contracts. We focus on analyzing the returns of three well-established delta-neutral option trading strategies: straddles, strangles, and butterfly spreads. First, we investigate the existence of cross-sectional momentum within the returns of these three strategies. Following this, we evaluate the persistence of the momentum anomaly, considering a spectrum of factors that are suggested to have predictive power within the options literature.

Our empirical analysis documents significant momentum effects in both straddles and strangles, while butterfly spreads exhibit no notable momentum phenomenon. The momentum patterns observed in straddles and strangles share similarities, particularly in their persistence over long-term horizons. Specifically, options with higher returns over the past one to three years consistently outperform those with lower returns over the same period when considering subsequent one-month returns. We find that a long/short portfolio based on returns from the past two years, excluding the most recent month, yields the most significant momentum returns for both strategies. Additionally, straddles exhibit a significant short-term reversal over a one-month horizon, denoting a tendency for the one-month performances to reverse in the subsequent month. Yet this reversal pattern is less prominent in the returns of strangles. Despite existing distinctions, the momentum patterns we observe in the options market are generally consistent with those found in the equity market. This includes a short-term reversal over the next month, suggested by (Jegadeesh (1990) and Lehmann (1990)), a medium-term momentum, found by (Jegadeesh and Titman (1993) and Jegadeesh and Titman (2001)), and a long-term reversal as documented by (De Bondt and Thaler (1985) and De Bondt and Thaler (1987)).

In our study of both straddles and strangles, robust momentum profits are observed even after controlling for a set of factors that are suggested to have predictive abilities in the options

literature.¹ These factors comprise the volatility spread, the idiosyncratic volatility, the market capitalization of the underlying stock, the term spread of implied volatility, and the smirk slope of implied volatility. Despite adjusting for these factors, the momentum portfolios continue to demonstrate significant alphas, thereby ruling out the possibility that momentum profits stem purely from exposure to those factors. Moreover, we investigate the explanatory ability of past returns in individual options via two-pass regressions and double sorts. Our findings suggest that returns observed over the previous one to three years significantly explain the pattern of risk-adjusted options returns in the subsequent month. This outcome aligns with the findings derived from our portfolio analyses, confirming a consistent pattern observed across different methods of evaluation.

Our study contributes to the existing body of momentum literature. While extensive research has confirmed the existence of momentum profits across diverse markets and asset classes (see, for example, Asness et al. (2013)), the exploration of momentum phenomena within the options markets has received comparatively less attention. Our systematic analysis bridges this gap by presenting strong evidence of the existence of momentum in the options market. Additionally, we observe even more pronounced momentum effects after controlling for conventional risk factors. Regarding the momentum patterns, unlike equity momentum, characterized by short-term reversals, medium-term momentum and longer-term reversals as documented in previous literature (Jegadeesh (1990), Jegadeesh and Titman (1993), and De Bondt and Thaler (1985)), our study identifies the absence of long-term reversals and limited evidence of short-term reversals in the options context. This discovery not only extends the understanding of momentum effects but also underscores the distinctive behavior of options momentum in comparison to equities.

Our study is closely related to Heston et al. (2023), which investigates options momentum returns on straddles across a broad spectrum of U.S. equities. However, the significant momentum returns observed by Heston et al. (2023) may potentially be driven by stock liquidity and might be confined to the straddle options strategy alone. Our study extends upon this by exploring a wider range of options strategies on individual equities characterized by more homogeneous liquidity features. We document strong momentum returns in both straddle and strangle strategies but not in the butterfly spreads. As evident in the straddles and strangles, options strategies betting on higher volatility of the underlying stocks, with no restriction on strategy payoffs, tend to exhibit

¹See Goyal and Saretto (2009), Cao and Han (2013), Zhan et al. (2022), Vasquez (2017), and Bali and Murray (2013).

momentum effects. Conversely, options strategies like butterfly spreads, betting on lower volatility in the underlying stocks and coupling with a capped payoff in the event of volatility shocks, are less likely to have momentum effects. This observation provides valuable implications for the relationship between options strategy characteristics and their propensity to generate momentum effects.

In the strand of equity options literature, prior studies have substantiated several predictors for the returns on individual equity options. Goyal and Saretto (2009) demonstrate the predictability of the volatility spread between implied and historical volatility. Concurrently, Cao and Han (2013) suggest that the idiosyncratic volatility of the underlying stock is a negative predictor of the cross-sectional option returns. Additionally, Zhan et al. (2022) highlight the predictability of market capitalization of the underlying stock. A recent study by Bali et al. (2023) finds that nonlinear machine learning models incorporating both option-based and stock-based characteristics provide significant options return predictability. Our contribution to the literature further strengthens this body of knowledge by confirming the predictive ability of momentum in the cross-section of individual options strategy returns across diverse options strategies.

The rest of this paper is organized as follows. The next section describes the data and the three option trading strategies. Section 3 presents the main results of our empirical studies. Section 4 discusses the robustness checks. Section 5 concludes.

2 Data and options trading strategies

We obtain option data from the OptionMetrics database. The data include bid-ask quotes, greeks, and other relevant options information on the S&P 100 constituent stocks from January 2000 to December 2019. The stock prices, returns, and market capitalization data are obtained from the CRSP database.

We construct combinations of options based on the trading strategies for each underlying stock on a monthly basis. On each expiration day, we select combinations of puts and calls that mature in the next month with the same expiration date. After the combination expires, we select a new combination of options that expire in the following month using the same criteria. We focus on three popular trading strategies that involve taking a position in a combination of options on the same stock. The three strategies are straddles, strangles, and butterfly spreads.

A straddle involves the simultaneous purchase of both a call and a put option, which share the

same strike prices and expiration dates. The strike price is close to the current stock price.² The straddle yields a profit when the underlying stock price experiences a sufficiently large movement in either direction before the expiration of the options.³

To form a straddle, we select a put/call pair with the same strike that is closest to at-the-money. Similar to Heston et al. (2023), we require the options to have positive open interest on the day they are selected and discard the pair if the call delta is greater than 0.75 or less than 0.25. We construct a zero-delta straddle for each pair by proportionally setting the put and call option weights to the absolute value of the delta of the other option, with a total weight of one. We hold the pairs to maturity. The return of a straddle is calculated by dividing the total profit of the call and put by the total cost. The profit equals the payoff minus the cost. The payoff of an option depends on the price of the underlying stock at expiration and the option's strike. The cost is set to be the midpoint between the bid and ask on the day they are selected.

A strangle involves the purchase of both a call and a put option that share the same expiration date yet have distinct strike prices. The strike price of the call is greater than that of the put.⁴ Like a straddle, a strangle makes a profit when the stock price makes a sufficiently large move in either direction, but loses all premiums if it expires between the two strikes.⁵ The profit pattern depends on the distance between the strike prices of the put and the call. The greater the distance, the further the stock price must move to make a profit. Compared to a straddle, the profitability of a strangle relies on greater price movement. However, when the underlying stock price ends up close to the strike price, the potential loss is comparatively lower when using a strangle.

To form a strangle, we select a put/call pair with strikes that are out-of-the-money and next to the one closest to at-the-money. Like the straddles, we require the at-the-money call delta to be no greater than 0.75 or less than 0.25, otherwise the pair is discarded. For each pair, we construct a zero-delta strangle by setting the put and call option weights to be proportional to the absolute value of the other one's delta and to sum to one. The strangles are assumed to be held to expiration. As with a straddle, the return of a strangle is calculated by dividing the total profit of the call and put by the total cost.

²In this paper, a straddle refers to a straddle purchase specifically. A straddle write involves selling a call and a put with the same strike price and expiration date, resulting in a reverse position. It is a highly risky strategy and is not discussed within the scope of this paper.

³A straddle is a suitable strategy for an investor who anticipates a substantial fluctuation in the stock price, yet lacks certainty regarding the direction of the movement.

⁴Within the context of this paper, the strangle strategy refers to a strangle purchase. The sale of a strangle is a risky strategy with an unlimited potential loss, thus it is not discussed here.

⁵By a strangle purchase, the investor is betting on a large move in the underlying stock price but is uncertain about its direction.

A butterfly spread involves options with three different strike prices; it can be created using either a group of call options or a group of put options. In this paper, we use call options to construct the strategy.⁶ This involves the purchase of a call with a comparatively lower strike price and another call with a relatively higher strike price, and the sale of two calls with a strike in the mid-point of the previous two strikes and close to the current underlying stock price. A butterfly yields a profit when the stock price ends up close to the middle strike price upon expiration, and loses all initial investments if the stock price expires outside of the strikes on either side.⁷ Typically, the initial investment of a butterfly spread is modest.

To create a butterfly spread, we select a combination of three calls with different strike prices: a lower strike, a higher strike, and a strike in the middle. The middle strike call is selected as the one with a strike closest to at-the-money. This criterion is identical to how we select a call for a straddle. Similarly, we require the call delta to fall between 0.75 and 0.25. Then, we select two calls with a lower strike and a higher strike that are closest to the middle strike on both sides. For each combination of options, we construct a butterfly spread by simply longing for one lower strike call and one higher strike call and shorting for two middle strike calls. We hold the options to expiration. The return of a butterfly is calculated as the total profit divided by the total cost. The total payoff is calculated as the difference between the payoffs of the two calls that go long and the two calls that go short. The total cost is the net cost of the options and is sometimes close to zero. To enhance the fairness of return calculation, we discard butterfly spreads with costs less than 0.001.

Thus, our final sample includes 19818 monthly pairs of put and call combinations for straddles, 14070 monthly combinations for strangles, and 12954 monthly combinations for butterfly spreads.

3 Empirical results

This section presents the main findings in our study of momentum in the three trading strategies. We begin with univariate sorts. Then we add controls for other option return predictors and examine whether the predictive ability of past returns remains by using time-series regression, Fama-MacBeth regression, and conditional double sorts.

⁶Using put options yields the same spread as using call options. The initial investment is the same in both cases, which can be demonstrated through put-call parity.

⁷It's a suitable strategy when an investor expects no big price movement prior to the expiry date.

3.1 Momentum in the cross-section

First, we investigate the existence of momentum in the cross-section using univariate sorts. For each month, we sort the underlying stocks into tertile portfolios⁸ based on the average returns⁹ during the formation period. We examine a variety of different formation periods ranging from one month to three years, with and without the most recent month. Table 1 reports the mean and t-statistic for the monthly portfolio returns of the three trading strategies. In the table, the numbers enclosed in parentheses in the first column labeled “Formation Period” represent the number of months preceding the current time t . For example, (1, 1) denotes the formation period from month $t-1$ to $t-1$, inclusively, equivalent to the return over the immediately preceding month. Similarly, (2, 36) indicates the period spanning month $t-2$ to $t-36$, encompassing a formation period of three years, with the exclusion of the most recent month, $t-1$. The return is computed based on the equal-weighted strategy returns of the underlying stocks for each tertile as well as the high-minus-low portfolio.

The empirical results in Table 1 suggest that both straddles and strangles exhibit significant momentum patterns; however, no clear momentum or reversal patterns can be identified for butterfly spreads. The analysis of straddles reveals three notable patterns. Firstly, a significant reversal at lag one is observed, as depicted in the first row of the table corresponding to the formation period (1, 1). The monthly portfolio return between the high and low positions attains an average monthly return of -5.82% (with a significance level of -3.95). This reversal can translate into an equivalent annualized return of 69.84%, substantially surpassing the documented short-term reversal observed in the equity market Medhat and Schmeling (2022). Due to the observed reversal pattern, the momentum portfolios based on the performance of the past 2, 6, and 12 months do not exhibit highly significant returns when the most recent month is included. However, upon comparing the monthly returns associated with the formation periods of (1, 12) and (2, 12), it becomes evident that the inclusion of the most recent month leads to a deterioration in monthly returns, decreasing from 5.845% to 3.910%. This finding underscores the second pattern: the emergence of a strong momentum effect in portfolios with formation periods ranging from one to three years when excluding the most recent month. This is evident in the rows corresponding to formation periods (2, 12), (2, 24), and (2, 36). Among these three portfolios, the one with a formation period comprising

⁸Due to the limited number of stocks in our dataset, we use tertile rather than quartile or quintile.

⁹Given the high volatility of option strategy returns, we use average return rather than compounded return to decrease the impact of the -100% returns when the options expire out-of-the-money.

the past two years when excluding the most recent month, denoted as (2, 24), exhibits the highest average monthly return of 7.711% (with a significance level of 4.95), resulting in an equivalent annual return of 92.53%. This (2, 24) portfolio also demonstrates stronger returns when compared to all univariate sorted outcomes observed within options associated with the entire universe of stocks in the U.S. market, as documented in Heston et al. (2023). Lastly, as the exclusion window expands to encompass one or two years, irrespective of whether the formation period extends to two or three years, the average monthly returns exhibit reduced significance. This phenomenon is evident in the performance of portfolios with formation periods of (13, 24), (13, 36), and (25, 36). While the formation period spanning $t-13$ to $t-24$ months still exhibits statistical significance, though with a significance level less than 3.0, the formation periods of $t-13$ to $t-36$ and $t-25$ to $t-36$ months do not reach statistical significance.

The momentum portfolio returns generated by the strangles strategy in Table 1 exhibit patterns that are identical to those observed in the straddles strategy with minor distinctions. The momentum remains evident in the tertile portfolio returns, though with somewhat less consistent monotonicity compared to straddles. Nevertheless, the short-term reversal is not significantly prominent. Similar to the straddles strategy, our findings indicate that, first, portfolios with formation periods encompassing the past one, two, and three years, while excluding the most recent month, consistently exhibit significant monthly returns of 11.272%, 16.984%, and 17.267% respectively, with corresponding t -statistics of 2.77, 3.94, and 3.52. Among these, the portfolio with a formation period of the past three years when excluding the most recent month, denoted as (2, 36), presents the highest monthly return of 17.267%, with a significance level of 3.52, underscoring its strength in generating returns. It is noteworthy that these three momentum portfolios yield stronger average returns in comparison to the results of straddles presented in Table 1 and the straddle findings reported in Heston et al. (2023). Furthermore, due to the absence of the short-term reversal effect, the momentum portfolio with a formation period spanning the past year, including the most recent month, demonstrates an even more substantial monthly return of 13.466% ($t=3.41$) compared to not including the most recent month. Similarly, the portfolio with a formation period extending from $t-13$ to $t-24$ also exhibits a significant monthly return of 14.613% ($t=3.22$). In contrast to the straddle strategy, the portfolio returns generated by the strangle strategy are frequently observed to be negative. This observation can in part be attributed to the inherent nature of the strangle strategy, which involves betting on a more volatile underlying price. Consequently, this leads to the occurrence of -100% returns when the options expire out-of-the-money.

Regarding butterfly spreads, our findings present an entirely different picture. Notably, only the formation period of (1, 1) demonstrates a marginally significant monthly return of -10.117% ($t=-2.22$). This finding suggests a weak short-term reversal pattern, resulting in an equivalent annual return of 121.404%. However, there is no observable indication of significant momentum patterns in the returns. The high-minus-low portfolio returns fail to exhibit statistical significance across all formation periods, and in general, statistical significance in tertile portfolio returns remains absent.

In summary, our findings regarding short-term reversal and long-term momentum in straddles and strangles are consistent with prior research in the stock literature (see, for example, Jegadeesh and Titman (1993) and Jegadeesh (1990)) and the study of straddle returns in a broader sample set by Heston et al. (2023). The formation period of the past two years, excluding the most recent month, offers the most significant momentum returns in both strategies. Expanded formation periods beyond two years exhibit less significant momentum returns. This result diverges from the findings of Heston et al. (2023) about straddles, in which the momentum return is most significantly indicated by the one-year lag and longer formation periods continue to offer significant high-minus-low returns. The potential reason may stem from our different data sets - we use a reduced data set consisting of the options on S&P 100 components, which are typically the most liquid optionable stocks, rather than the entire optionable stocks in the U.S. market.

To further examine and compare the momentum returns, we present their average monthly returns alongside the corresponding t -statistics for the three momentum strategies in Figure 1. In the figure, the blue bars represent the monthly returns, while the orange bars denote the associated t -statistics. As depicted in the figure, it is evident that the returns associated with both the straddle and strangle strategies exhibit an inverted-U shape. The highest returns, and the most statistically significant ones, consistently originate from portfolios with formation periods spanning one, two, and three years, while excluding the most recent month. In contrast, the returns linked to other formation periods show a declining trend. This distinctive pattern diverges from the equity market, where momentum is primarily driven by the performance of the past seven to twelve months, as documented by Novy-Marx (2012). Our findings reveal that within the options market, strategies based on past horizons of one year (intermediate), two years (long-term), and three years (very long-term), while excluding the most recent month, are the drivers of momentum. This pattern is particularly pronounced among options written on the largest and most liquid stocks, such as those represented by the constituent firms in the S&P 100 Index.

Table 1: **Univariate Sorts of Returns generated by Straddle, Strangle, and Butterfly Strategies.** This table reports the means and t -statistics for the monthly returns of three portfolios, straddle, strangle and butterfly. The underlying stocks are sorted into tertile portfolios each month based on the average returns during the formation period shown in the row header. The numbers enclosed in parentheses denote the number of months preceding the current time t . To clarify, for instance, (1,1) signifies the formation period spanning from $t-1$ to $t-1$, inclusively, equivalent to the return over the immediately preceding month; (2,36) indicates the period spanning from $t-2$ to $t-36$, encompassing a formation period of three years, excluding the most recent month, $t-1$. The return is computed based on the equal-weighted strategy returns of the underlying stocks. The returns are expressed in percentages, with, for example, “7.711” denoting an average monthly return of 7.711%. The t -statistics in parentheses are corrected using Newey-West standard errors with three lags. The returns and t -statistics highlighted in bold are statistically significant at the 1% level, while the results in italics hold significance at the 2.5% level.

Formation	Straddle				Strangle				Butterfly			
Period	Low	Medium	High	H-L	Low	Medium	High	H-L	Low	Medium	High	H-L
(1,1)	4.883	1.788	-0.939	-5.821	-25.452	-22.794	-25.847	-0.395	2.152	-11.163	-7.965	<i>-10.117</i>
	(2.41)	(0.96)	(-0.48)	(-3.95)	(-5.96)	(-4.81)	(-6.75)	(-0.12)	(0.56)	(-3.18)	(-2.49)	<i>(-2.22)</i>
(1,2)	3.623	2.462	-0.913	-4.536	-25.953	-24.542	-25.543	0.41	-0.784	6.841	-7.69	-6.907
	(1.73)	(1.34)	(-0.48)	(-2.99)	(-5.5)	(-5.15)	(-7.04)	(0.11)	(-0.24)	(-1.79)	(-2.33)	(-1.64)
(1,6)	1.2	-0.344	5.116	3.916	-29.441	-25.887	-21.602	7.84	2.547	8.242	-5.935	3.388
	(0.59)	(-0.18)	(2.61)	(2.48)	(-6.68)	(-5.87)	(-5.4)	(1.82)	(-0.76)	(-2.56)	(-1.72)	(-0.78)
(1,12)	0.888	0.088	4.798	<i>3.91</i>	-30.731	-30.994	-17.265	13.466	6.105	-6.977	-4.703	1.402
	(0.45)	(0.04)	(2.32)	<i>(2.27)</i>	(-6.36)	(-7.25)	(-4.47)	(3.41)	(-2.08)	(-2.05)	(-1.49)	(0.37)
(2,12)	-0.213	0.368	5.632	5.845	-28.934	-32.397	-17.662	11.272	7.253	5.511	-4.512	2.741
	(-0.11)	(0.19)	(2.66)	(3.64)	(-5.83)	(-8.16)	(-4.61)	(2.77)	(-2.27)	(-1.71)	(-1.21)	(0.7)
(2,24)	-2.116	2.569	5.594	7.711	-33.738	-27.701	-16.753	16.984	-5.377	-7.683	5.218	0.158
	(-1.11)	(1.24)	(2.69)	(4.95)	(-7.68)	(-5.76)	(-3.64)	(3.94)	(-1.6)	(-2.55)	(-1.58)	(0.05)
(2,36)	-0.461	1.097	5.335	5.795	-33.857	-29.258	-16.589	17.267	-8.6	-4.02	-6.573	2.026
	(-0.25)	(0.54)	(2.48)	(3.52)	(-7.78)	(-6.6)	(-3.26)	(3.52)	(-2.71)	(-1.31)	(-1.77)	(0.5)
(13,24)	-0.549	2.351	3.873	4.422	-36.029	-20.921	-21.416	14.613	-5.404	-4.32	-7.83	-2.426
	(-0.28)	(1.11)	(1.87)	(2.81)	(-8.36)	(-3.99)	(-4.71)	(3.22)	(-1.69)	(-1.29)	(-2.48)	(-0.85)
(13,36)	0.232	1.905	3.423	3.191	-30.507	-29.355	-18.827	<i>11.68</i>	6.841	5.814	5.961	0.88
	(0.13)	(0.9)	(1.61)	(1.81)	(-7.27)	(-6.35)	(-3.66)	<i>(2.18)</i>	(-2.33)	(-1.78)	(-1.7)	(0.25)
(25,36)	2.031	0.956	2.358	0.327	-33.107	-24.963	-21.561	<i>11.546</i>	-5.774	-7.544	-5.83	-0.056
	(0.99)	(0.49)	(1.16)	(0.21)	(-7.87)	(-5.7)	(-4.08)	<i>(2.29)</i>	(-1.86)	(-2.38)	(-1.48)	(-0.01)

3.2 Control for other predictors

Our next task is to study the momentum pattern after adding controls for other option return predictors. We examine the risk-adjusted returns of momentum by time-series regression and explore the relationship between the predictors and individual option strategy returns by using Fama-MacBeth regression and conditional double sorts.

Given the absence of a general formal theoretical model for the cross-section of option returns, we follow Heston et al. (2023) and consider five predictors that the options literature suggests to have predictive abilities. These are the volatility spread, idiosyncratic volatility, the market capitalization of the underlying stock, the term spread of implied volatility, and the smirk slope of implied volatility.

The volatility spread is calculated as the difference between implied and historical volatility (see Goyal and Saretto (2009)). The implied volatility is computed using the average of the one-month at-the-money put and call implied volatility, and the historical volatility is computed using

Figure 1: This figure illustrates the monthly average returns with the t statistics of momentum portfolios generated using the strategies of (a) straddle; (b) strangle; and (c) butterfly, across a variety of formation periods.



the standard deviation of realized daily stock returns across the most recent 12 months. The idiosyncratic volatility is calculated using the standard deviation of the residuals of the Fama-French three-factor model (see Cao and Han (2013)). The residuals are estimated using the daily returns over a 22-day rolling regression. The market capitalization is simply that of the underlying stock (see Zhan et al. (2022)). The term spread of implied volatility is computed using the difference between the long-term and the short-term at-the-money implied volatility (see Vasquez (2017)). The short-term volatility is defined as the average of the one-month at-the-money put implied volatility and call implied volatility, and the long-term volatility is defined as the 60-day average implied volatility. The smirk is measured as the slope of the one-month implied volatility curve and is computed using the difference between the implied volatility of the one-month call with a delta closest to 0.3 and the one-month put with a delta closest to -0.3. This is related to the study of Bali and Murray (2013) and we follow the computation of Heston et al. (2023) for simplicity.

3.2.1 Risk-adjusted returns

We first investigate whether the profitability of momentum can be explained by other factors that are already in the literature. We regress the momentum returns of the three trading strategies on a linear pricing model consisting of the five aforementioned factors:

$$R_t^p = \alpha_p + \beta_p' F_t + e_t^p, \quad t = 1, 2, \dots, T, \quad (1)$$

for each strategy, where R^p is the momentum return (i.e., a long/short portfolio return) of an options trading strategy and F is the return of the factors. Since all the factors are long/short portfolios, the intercept α_p can be interpreted as a measure of mispricing with respect to the factor model. Tables 2, 3, and 4 report the results of these regressions for the three trading strategies. We consider a series of momentum formation periods.

In Table 2, the regression results of straddle strategies show three interesting patterns. Firstly, the short-term reversal in the first row with a formation period of the past one month, denoted as (1, 1) displays a significant momentum alpha of -5.092 with $t=-2.68$. None of the five factors significantly explains the reversal return. This finding diverges from Heston et al. (2023), where short-term reversals can be explained by the difference between implied and historical volatility (referred to as “IV-HV”) and the term spread of implied volatility (referred to as “TermSpread”). Secondly, the formation periods that skip one month (i.e., the formation periods of the past (2, 12), (2, 24), and (2, 36) months) yield the most significant momentum alphas of 7.244 ($t=3.59$), 9.267 ($t=4.43$), and 10.103 ($t=4.80$) respectively. These three momentum returns are significantly explained by the idiosyncratic volatility (Cao and Han (2013)), referred to as “IdiosynVol”. This finding is in line with the results reported by Heston et al. (2023), where the idiosyncratic volatility does explain some of the fluctuation in momentum returns for the formation period of (2, 12). The smirk slope of implied volatility (Bali and Murray (2013)), referred to as “SmirkSlope”, also significantly explains the momentum returns for formation periods of (2, 12) and (2, 24), and marginally explains other momentum returns. Lastly, the other three factors - the volatility spread (“IV-HV”), the market capitalization of the underlying stock (“MktCap”), and the term spread of implied volatility (“TermSpread”) - demonstrate limited explanatory power in relation to the momentum returns across almost all formation periods. This result is notably different from those results presented in Heston et al. (2023), which proves the significant explanatory power of these three factors. Heston’s results indicate the significance of market capitalization by incorporating

both large- and small-cap stocks within the entire universe. In contrast, our findings align with our expectations, as our dataset exclusively comprises large-cap stocks, inherently excluding the market-cap effect. Nonetheless, the exclusion of the market-cap effect does not negate the impact of the volatility spread (“IV-HV”) and the term spread of implied volatility (“TermSpread”). Consequently, we argue that the explanatory power of these two factors for both the short-term reversal and momentum is likely influenced by the characteristics of less liquid optionable stocks. Overall, it is evident that both the reversal and momentum profits do not simply emerge because of exposure to other factors.

Table 2: Risk-adjusted straddle momentum returns. This table reports the results from time-series regressions of straddle momentum returns on a linear pricing model consisting of the five factors described in section 3.2. The momentum returns are computed based on a series of formation periods. Each row presents the regression results of a particular formation period given in the first column. The t -statistics in parentheses are corrected using Newey-West standard errors with three lags.

Formation period	Straddles						R^2
	Intercept	IV-HV	IdiosynVol	MktCap	TermSpread	SmirkSlope	
(1, 1)	-5.092 (-2.68)	0.009 (0.11)	0.102 (1.26)	0.052 (0.58)	-0.111 (-1.56)	0.017 (0.23)	0.025
(1, 2)	-6.246 (-3.33)	-0.028 (-0.45)	0.106 (1.47)	0.077 (0.93)	-0.061 (-0.82)	-0.132 (-1.65)	0.036
(1, 6)	3.587 (1.72)	-0.056 (-0.65)	0.220 (2.81)	-0.003 (-0.03)	-0.098 (-1.26)	-0.173 (-2.21)	0.100
(1, 12)	5.224 (2.69)	-0.009 (-0.11)	0.308 (4.53)	0.033 (0.34)	-0.072 (-0.94)	-0.131 (-2.26)	0.162
(2, 12)	7.244 (3.59)	-0.004 (-0.06)	0.320 (4.62)	0.025 (0.28)	0.018 (0.22)	-0.196 (-2.94)	0.182
(2, 24)	9.267 (4.43)	-0.040 (-0.38)	0.308 (3.37)	0.033 (0.37)	0.028 (0.34)	-0.240 (-3.43)	0.175
(2, 36)	10.103 (4.80)	0.080 (0.92)	0.246 (3.73)	0.055 (0.65)	0.012 (0.14)	-0.120 (-1.66)	0.114
(13, 24)	5.152 (2.33)	-0.015 (-0.15)	0.141 (1.42)	-0.034 (-0.41)	0.061 (0.77)	-0.046 (-0.53)	0.039
(13, 36)	4.856 (2.08)	0.106 (1.21)	0.077 (0.93)	0.072 (1.12)	-0.037 (-0.48)	0.062 (0.64)	0.029
(25, 36)	1.413 (0.71)	0.313 (5.31)	-0.062 (-0.98)	0.093 (1.20)	-0.008 (-0.13)	0.112 (1.57)	0.142

Note: In the table, “IV-HV” designates the volatility spread; “IdiosynVol” corresponds to the idiosyncratic volatility; “MktCap” represents the market capitalization of the underlying stock; “TermSpread” pertains to the term spread of implied volatility; and “SmirkSlope” denotes the smirk slope of implied volatility.

Table 3 shows the regression results for strangle strategies. Similar to straddles, the skip-one-month formation periods spanning one, two and three years (denoted as (2, 12), (2, 24), (2, 36)) yield the most significant momentum alphas for strangles. These alphas reach exceptionally high levels with monthly returns of 15.06% ($t=3.24$), 20.89% ($t=3.70$) and 23.69% ($t=3.86$), respectively. These figures substantially surpass the momentum alphas generated by the straddle strategy, which

typically fall within the range of 7%-10%. The idiosyncratic volatility (“IdiosynVol”) remains a significant factor in explaining the variations in momentum returns with formation periods of (1, 6), (1, 12), (2, 12), and (2, 24), but the smirk slope (“SmirkSlope”) plays a less part. The momentum returns based on longer “skip windows”, such as (13, 24), (13, 36), and (25, 36), can hardly be explained by any of the factors and generate less significant alphas. In addition, the short-term reversals for strangles do not exhibit any significant alpha, and none of the five factors show significant explanatory power. This is consistent with the results presented in Table 1, which indicates that short-term reversals do not produce significant high-minus-low returns.

Table 3: Risk-adjusted strangle momentum returns. This table reports the results from time-series regressions of strangle momentum returns on a linear pricing model consisting of the five factors described in section 3.2. The momentum returns are computed based on a series of formation periods. Each row presents the regression results of a particular formation period given in the first column. The t-statistics in parentheses are corrected using Newey-West standard errors with three lags.

Formation period	Strangles						R^2
	Intercept	IV-HV	IdiosynVol	MktCap	TermSpread	SmirkSlope	
(1, 1)	2.975 (0.69)	-0.084 (-1.04)	0.154 (1.31)	-0.035 (-0.46)	0.087 (1.00)	-0.118 (-1.75)	0.056
(1, 2)	0.614 (0.15)	0.049 (0.70)	0.186 (1.96)	0.042 (0.51)	0.093 (1.33)	-0.216 (-3.86)	0.176
(1, 6)	6.161 (1.37)	0.014 (0.17)	0.434 (4.96)	0.132 (1.23)	0.013 (0.17)	-0.073 (-0.97)	0.232
(1, 12)	11.720 (2.57)	-0.024 (-0.26)	0.415 (5.14)	0.142 (1.48)	0.042 (0.42)	-0.206 (-2.02)	0.231
(2, 12)	15.061 (3.24)	0.016 (0.20)	0.404 (5.43)	0.187 (2.19)	0.097 (1.10)	-0.149 (-1.78)	0.221
(2, 24)	20.899 (3.70)	0.117 (1.09)	0.327 (3.03)	-0.005 (-0.04)	0.089 (0.90)	-0.301 (-2.38)	0.250
(2, 36)	23.690 (3.86)	0.208 (1.92)	0.120 (1.01)	0.207 (1.44)	0.096 (0.75)	-0.166 (-1.05)	0.083
(13, 24)	15.642 (2.58)	0.033 (0.35)	0.003 (0.03)	-0.254 (-2.31)	0.005 (0.05)	-0.237 (-2.24)	0.106
(13, 36)	19.044 (2.95)	0.096 (0.97)	-0.014 (-0.13)	0.054 (0.50)	0.077 (0.66)	-0.134 (-1.10)	0.032
(25, 36)	9.337 (2.00)	0.142 (1.47)	0.116 (1.15)	0.297 (2.68)	0.061 (0.59)	-0.104 (-0.79)	0.102

Note: In the table, “IV-HV” designates the volatility spread; “IdiosynVol” corresponds to the idiosyncratic volatility; “MktCap” represents the market capitalization of the underlying stock; “TermSpread” pertains to the term spread of implied volatility; and “SmirkSlope” denotes the smirk slope of implied volatility.

The butterfly spreads strategy, as shown in Table 4, exhibits a markedly distinct pattern when compared to the results of straddle and strangle strategies. The returns display a weak reversal phenomenon within the formation period of (1, 1), resulting in a marginally significant alpha of -7.70% ($t=1.84$). Notably, the factors of volatility spread (referred to as “IV-HV”) and idiosyncratic volatility (referred to as “IdiosynVol”) provide a marginal level of significance in explaining the

momentum return, with corresponding t -statistics of -2.22 and -1.91 respectively. Other factors do not demonstrate any meaningful explanatory power. The returns generated by the butterfly strategy for formation periods other than (1, 1) can hardly be explained by any factors and fail to yield statistically significant alphas. This observation is consistent with the results presented in Table 1, where the high-minus-low butterfly spread returns lack statistical significance across all formation periods.

Table 4: Risk-adjusted butterfly spread momentum returns. This table reports the results from time-series regressions of butterfly spread momentum returns on a linear pricing model consisting of the five factors described in section 3.2. The momentum returns are computed based on a series of formation periods. Each row presents the regression results of a particular formation period given in the first column. The t -statistics in parentheses are corrected using Newey-West standard errors with three lags.

Formation period	Butterfly spreads						R^2
	Intercept	IV-HV	IdiosynVol	MktCap	TermSpread	SmirkSlope	
(1, 1)	-7.700 (-1.84)	-0.159 (-2.22)	-0.241 (-1.91)	-0.165 (-1.87)	-0.014 (-0.28)	-0.063 (-0.51)	0.085
(1, 2)	-4.494 (-1.00)	-0.149 (-1.49)	-0.153 (-1.27)	0.095 (0.89)	0.170 (1.67)	-0.079 (-0.77)	0.152
(1, 6)	-5.138 (-1.14)	-0.009 (-0.10)	-0.230 (-2.05)	0.031 (0.30)	0.096 (2.36)	-0.079 (-0.92)	0.101
(1, 12)	-0.844 (-0.20)	-0.064 (-0.55)	-0.260 (-2.04)	0.054 (0.46)	0.034 (0.71)	-0.021 (-0.17)	0.090
(2, 12)	0.147 (0.03)	-0.175 (-1.66)	-0.180 (-1.41)	0.107 (0.78)	0.055 (0.91)	-0.059 (-0.60)	0.097
(2, 24)	4.914 (1.24)	-0.320 (-3.90)	0.038 (0.38)	0.025 (0.27)	0.096 (1.02)	-0.347 (-3.59)	0.180
(2, 36)	-3.181 (-0.86)	-0.189 (-2.14)	0.020 (0.22)	0.234 (2.18)	0.178 (2.43)	-0.255 (-2.78)	0.219
(13, 24)	3.862 (1.03)	-0.166 (-1.55)	0.051 (0.53)	-0.185 (-1.43)	-0.063 (-0.48)	-0.203 (-2.19)	0.105
(13, 36)	-1.517 (-0.40)	-0.179 (-1.75)	0.170 (1.72)	0.110 (0.73)	0.072 (0.63)	-0.253 (-2.56)	0.098
(25, 36)	-9.254 (-2.01)	0.023 (0.18)	0.041 (0.39)	0.130 (0.81)	0.292 (3.70)	-0.110 (-0.72)	0.232

To summarize, for both straddles and strangles, the momentum returns based on the skip-one-month formation periods survive the factor adjustments and provide significant alphas. The momentum returns can partly be explained by one or two factors; however, it is evident that they are not simply a result of exposure to the five factors. Regarding butterfly spreads, the momentum alpha is not significant at all formation periods.

3.2.2 Cross-sectional regressions

Next, instead of examining the returns of momentum portfolios, we investigate the risk-adjusted returns of individual stock option strategies. We run cross-sectional regressions using the Fama-

MacBeth procedure. Under the null hypothesis, the risk-adjusted returns are independent of past returns. Thus, we examine whether the past return can significantly explain the returns of individual option strategies after controlling for the same factors as in section 3.2.1.

First, we find beta estimates for each individual option return through time-series regressions:

$$R_t^i = \alpha_i + \beta_i' F_t + e_t^i, \quad t = 1, 2, \dots, T, \quad (2)$$

where R^i represents the options strategy return of an individual underlying stock, and F denotes the returns of the factors, as described in section 3.2.1. We then run a cross-sectional regression of the risk-adjusted returns of individual options strategies on their respective historical performance. Our regression follows the specifications outlined by Brennan et al. (1998) and Goyal and Saretto (2009):

$$R_t^i - \hat{\beta}_i' F_t = a_t + \gamma_t z_t^i + \epsilon_t^i, \quad i = 1, 2, \dots, N, \quad (3)$$

where β_i represents the beta estimates derived from the first-pass time-series regression, and z^i denotes the historical returns. This cross-sectional regression is conducted monthly, and we estimate a and γ as the time-series average of the monthly estimates.

Table 5 reports the results of the cross-sectional regressions for the three trading strategies, considering a series of formation periods. It is evident that the past returns significantly explain the risk-adjusted returns of both straddle and strangle strategies. More specifically, within straddles, the past returns exhibit statistical significance in explaining the returns across relatively shorter formation periods ranging from the past one month (1, 1) to the past three years, denoted as (2, 36). The explanatory power of returns for the strangle strategies remains significant for longer formation periods, ranging from (1, 12) to (13, 36). This further demonstrates the existence of momentum effects in straddle and strangle returns. It is noteworthy that in alignment with time series regressions, past returns exhibit significance in explaining risk-adjusted short-term reversals - the one-month formation period (1, 1) - for straddle strategies. In contrast, the strangle strategy does not exhibit any short-term reversal, as evidenced by the findings presented in Table 1.

Remarkably, it is observed that the t -statistics associated with the intercept coefficients in straddle regressions consistently remain below 1 across all formation periods. In contrast, the t -statistics for strangle regressions exhibit high significance across all formation periods, with values exceeding 4.0 in all cases. This observation suggests that past returns can fully explain the risk-

adjusted returns for straddles within a linear framework. In contrast, this pattern does not hold true for strangles, indicating the existence of additional factors that influence the risk-adjusted returns of strangles in addition to past returns.

In the context of butterfly spreads, as depicted in Table 5, it is noteworthy that the t -statistics for regression coefficients of past returns are almost all insignificant. Conversely, nearly all intercepts exhibit marginal levels of significance for short and long formation periods, encompassing periods ranging from (1, 1) to (1, 6), as well as (2, 36) and (25, 36). These findings suggest that the risk-adjusted returns for butterfly spreads cannot be explained by past returns alone. Instead, they imply the presence of potential additional drivers influencing risk-adjusted returns in a linear structure for the specific formation periods.

Table 5: Cross-sectional regressions of risk-adjusted individual option returns. This table reports the results from cross-sectional regressions of risk-adjusted individual option returns on past returns using the Fama-MacBeth procedure. The individual option returns are strategy returns for individual underlying stocks based on straddles, strangles, and butterfly spreads, respectively. The past returns are computed based on a series of formation periods. Each row presents the regression results of a particular formation period given in the first column. The t -statistics in parentheses are corrected using Newey-West standard errors with three lags.

Formation period	Straddles			Strangles			Butterfly spreads		
	Intercept	PastReturn	$\bar{R}^2$	Intercept	PastReturn	$\bar{R}^2$	Intercept	PastReturn	$\bar{R}^2$
(1, 1)	1.262 (0.66)	-0.024 (-2.44)	0.018	-29.078 (-7.69)	0.011 (0.97)	0.018	-4.563 (-2.06)	-0.008 (-0.57)	0.022
(1, 2)	1.319 (0.69)	-0.031 (-2.65)	0.016	-29.266 (-7.70)	0.014 (1.02)	0.013	-4.076 (-1.81)	0.017 (0.99)	0.022
(1, 6)	1.621 (0.84)	0.065 (3.02)	0.018	-27.643 (-7.27)	0.067 (2.92)	0.019	-4.430 (-2.08)	0.037 (1.25)	0.026
(1, 12)	1.196 (0.63)	0.092 (3.34)	0.018	-23.958 (-6.11)	0.155 (5.12)	0.021	-3.815 (-1.75)	0.054 (1.44)	0.024
(2, 12)	1.145 (0.61)	0.107 (4.04)	0.018	-24.602 (-6.27)	0.142 (4.84)	0.021	-3.728 (-1.71)	0.062 (1.72)	0.022
(2, 24)	1.040 (0.56)	0.145 (3.96)	0.018	-22.123 (-5.14)	0.197 (5.00)	0.023	-3.871 (-1.74)	0.112 (2.00)	0.026
(2, 36)	1.409 (0.74)	0.180 (4.17)	0.017	-20.781 (-4.38)	0.236 (4.57)	0.025	-4.412 (-2.02)	0.025 (0.36)	0.023
(13, 24)	1.325 (0.71)	0.051 (1.94)	0.016	-25.568 (-5.93)	0.089 (2.84)	0.016	-4.645 (-2.12)	0.046 (1.33)	0.019
(13, 36)	1.617 (0.85)	0.067 (1.85)	0.017	-24.045 (-5.14)	0.139 (3.51)	0.017	-4.342 (-1.92)	-0.012 (-0.23)	0.019
(25, 36)	1.283 (0.68)	0.026 (1.02)	0.015	-26.119 (-5.77)	0.086 (2.72)	0.016	-4.679 (-2.03)	-0.064 (-1.48)	0.020

3.2.3 Conditional two-way sorts

We proceed with an examination of conditional two-way sorts, seeking to offer an alternative perspective on the existence of momentum while refraining from imposing a linear structure. We focus on the formation period spanning the prior 2 to 24 months, denoted as (2, 24). This formation period has been consistently identified as exhibiting the most substantial momentum effect, as evident from prior empirical studies across Tables 1 to 5.

In this study, we continue employing the same set of five control variables, as explored in Section 3.2.1. These variables encompass the volatility spread (“IV-HV”), the idiosyncratic volatility (“IdiosynVol”), the market capitalization of the underlying stock (“MktCap”), the term spread of implied volatility (“TermSpread”), and the smirk slope of implied volatility (“SmirkSlope”).¹⁰

For each month, we select one of the five control variables and sort stocks into tertiles based on this chosen variable. Subsequently, within each tertile, we further sort the stocks, once again into tertiles, based on the average returns of the strategy over the formation period. We then proceed to construct an equal-weighted portfolio: for each tertile generated based on past returns of the strategy portfolio, we compute the average of the portfolio returns over the tertiles created using the initially selected control variable. In essence, we calculate the average return of the past-return-sorted tertile portfolio returns across the three levels of the selected control variable. This approach allows us to assess the momentum effect while controlling for the specific variable without imposing a linear structure. We term this portfolio as one within a “neutral” context related to the selected control variable. This is because the returns are aggregated across the three tertiles spanning “Low” to “High” within the chosen control variable. Consequently, this aggregation effectively eliminates the influence of the control variable itself. Thus, we evaluate the momentum effect of options strategies within a neutral context defined by each of the chosen control variables. For instance, when we select the volatility spread (“IV-HV”) as the control variable, we examine the momentum effect within the volatility-spread-neutral context. We repeat this evaluation process for each of the five control variables, systematically investigating the momentum effect in neutral contexts associated with each variable.

Table 6 reports the monthly means and associated t -statistics of tertile returns in conditional two-way sorts for the three trading strategies. The “High-Low” returns represent the momentum returns within the neutral context specific to each corresponding variable. It is important to note that, the double-sort methodology allows for the control of only a single variable at a time.

¹⁰For detailed information on the computation of these five variables, please refer to Section 3.2.1

Therefore, each row in Table 6 presents the results associated with a single control variable. A significantly positive (negative) mean value for the “High-Low” returns represents a strong indicator of the existence of a significant momentum (reversal) return within the neutral context corresponding to the variable shown in the row header.

The returns presented in Table 6 reveal three interesting patterns. Firstly, within the neutral context associated with certain control variables, all three strategies produce returns that are not only stronger but also exhibit a higher level of statistical significance, when compared to those obtained through unconditional univariate sorts, as presented in Table 1. For example, the idiosyncratic-volatility-neutral straddle strategy (indicated in the row labeled “IdiosynVol”) delivers the highest monthly return of 13.363% ($t=5.32$). In contrast, as shown in Table 1, the original straddle strategy with a formation period of (2, 24) yields a significantly lower monthly return of 7.711% ($t=4.95$). Similarly, the IV-term-spread-neutral straddle strategy (designated in the row for “TermSpread”) achieves a notable monthly return of 13.067% ($t=4.64$). Even the third performing volatility-spread-neutral straddle strategy (designated in the row for “IV-HV”) attains a monthly return of 12.298% ($t=4.87$). In the strangle strategies, we observe similar patterns. The IV-term-spread-neutral strangle strategy (designated in the row for “TermSpread”) contributes the highest monthly return of 37.679% ($t=5.65$), which shows a significantly higher return than the return of 16.981% ($t=3.94$), generated by the original strangle strategy with a formation period of (2, 24), as presented in Table 1. Similarly, the volatility-spread-neutral strangle strategy (designated in the row for “IV-HV”) also attains a notable monthly return of 28.90% ($t=4.34$). Notably, while the momentum returns of the butterfly spreads remain statistically insignificant, both the magnitude and the associated significance levels of the monthly returns are markedly higher than those generated by the original butterfly spreads strategy. For instance, the smirk-slope-neutral butterfly spreads achieves a monthly return of 6.508% ($t=1.05$); in contrast, the corresponding original butterfly spreads yield a monthly return of just 0.158% ($t=0.05$). This suggests that a more pronounced (though still statistically insignificant) momentum pattern becomes evident within the neutral context of specific control variables. This observation aligns with the findings for the straddle and strangle strategies.

The patterns in the neutral context suggest that the five selected variables may not make a significant contribution to the profitability of the options strategy momentum. Instead, their presence deteriorates the momentum patterns, particularly in the case of strategies of idiosyncratic-volatility-neutral straddle and IV-term-spread-neutral strangle. When we account for the effect of

idiosyncratic volatility and the term spread respectively, the corresponding momentum strategies achieve a significantly higher return.

Secondly, when comparing the returns of straddle and strangle strategies in Table 6, we observe that the long legs (as indicated in the “High” column of the table) within the straddle strategy yield significantly higher magnitude returns than the short legs (as indicated in the “Short” column of the table). The returns of long legs generally range from 5.12%-5.89% and are twice those of short legs, which are around -1.11% to 2.03%. This demonstrates that the long legs primarily contribute to the strategy returns in straddles. In contrast, strangle returns show entirely opposite results, where the short legs yield monthly returns of around 30.51% to 37.54%, while the long legs achieve monthly returns ranging from 14.93% to 17.86%. Hence, the short legs contribute to the strategy returns in strangles.

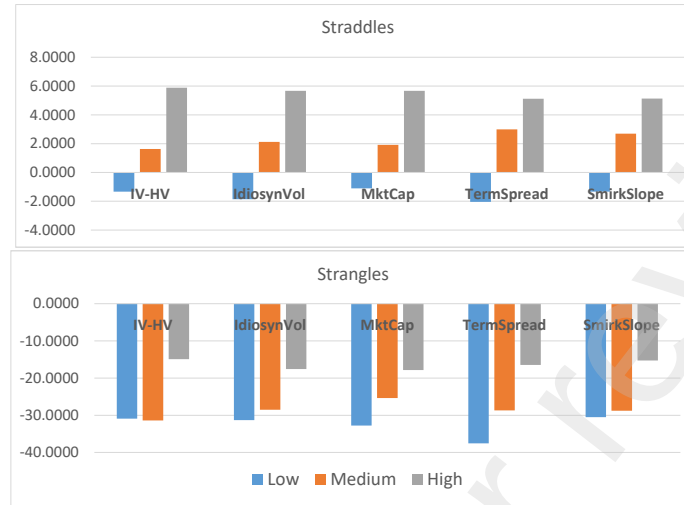
Lastly, we discern a distinct pattern of monotonicity in the returns of the tertile portfolios for both straddle and strangle strategies within the neutral context associated with each of the control variables, as depicted in Table 6. This monotonic relationship is further visualized in Figure 2. It is evident that the returns of the tertile portfolios consistently increase from “Low” to “Medium” and ultimately reach their highest levels in the “High” category. This observation strongly indicates the presence of clear momentum patterns in the neutral context associated with each of the five control factors.

Table 6: Double sorts at formation period of lags 2 to 24. This table reports the results of conditional two-way sorts for the three strategies. For each month, we first sort stocks into tertiles based on a control variable shown in the row header. After that, within each tertile, we sort the stocks, again into tertiles, based on the strategy’s average returns during the formation period of the past 2-to-24 months and construct an equal-weighted portfolio. We then take the average of the past-return-based portfolio returns over the control variable tertiles. Each row presents the means and t-statistics, in parentheses, of the past-return-sorted tertile portfolio returns after controlling for the corresponding control variable. The t-statistics are corrected using Newey-West standard errors with three lags.

Control	Straddles				Strangles				Butterfly spreads			
	Low	Medium	High	High-Low	Low	Medium	High	High-Low	Low	Medium	High	High-Low
IV-HV	-1.333 (-0.70)	1.636 (0.79)	5.890 (2.90)	12.298 (4.87)	-30.914 (-6.99)	-31.412 (-7.49)	-14.927 (-3.05)	28.900 (4.34)	-7.424 (-2.09)	-6.617 (-2.08)	-4.641 (-1.65)	4.575 (0.85)
IdiosynVol	-1.857 (-1.01)	2.128 (1.09)	5.672 (2.85)	13.363 (5.32)	-31.307 (-7.96)	-28.510 (-6.77)	-17.576 (-3.73)	24.882 (3.79)	-5.722 (-2.09)	-6.834 (-2.43)	-3.124 (-0.95)	5.147 (0.84)
MktCap	-1.105 (-0.59)	1.914 (0.98)	5.671 (2.89)	11.652 (4.40)	-32.758 (-6.69)	-25.370 (-5.76)	-17.860 (-4.19)	24.944 (3.65)	-6.436 (-1.81)	-5.457 (-2.01)	-5.433 (-1.97)	2.423 (0.42)
TermSpread	-2.039 (-1.10)	2.993 (1.43)	5.116 (2.59)	13.067 (4.64)	-37.544 (-7.64)	-28.682 (-6.01)	-16.472 (-3.89)	37.679 (5.65)	-7.528 (-2.93)	-3.435 (-0.92)	-4.938 (-1.79)	2.479 (0.43)
SmirkSlope	-1.330 (-0.74)	2.694 (1.35)	5.128 (2.51)	10.363 (3.87)	-30.508 (-7.48)	-28.771 (-6.43)	-15.263 (-3.43)	23.400 (3.28)	-6.201 (-1.86)	-8.999 (-3.42)	-2.041 (-0.54)	6.508 (1.05)

Note: In the table, “IV-HV” designates the volatility spread; “IdiosynVol” corresponds to the idiosyncratic volatility; “MktCap” represents the market capitalization of the underlying stock; “TermSpread” pertains to the term spread of implied volatility; and “SmirkSlope” denotes the smirk slope of implied volatility.

Figure 2: Monotonicity. This figure illustrates the monthly average returns of momentum portfolios generated using the strategies of straddles and strangles with the formation periods of (2, 24) in Table 6. The Low, Medium, and High correspond to the returns of tertiles portfolio in a neutral context associated with five control variables (see the details of those variables in Table 6).



In summary, our analysis in Sections 3.2.1, 3.2.2, and 3.2.3 investigates the momentum effects within three options trading strategies while accounting for five other factors that have been recognized as having predictive power in the options literature. Our assessment of the momentum pattern includes an analysis of both the long/short portfolio and individual options. With respect to straddle and strangle returns, we observe consistent momentum effects even after controlling for these predictive variables. In contrast, for butterfly spread returns, we do not find any significant momentum returns or predictive capabilities associated with the control variables. Remarkably, our findings demonstrate even more pronounced momentum patterns and returns within the neutral context associated with idiosyncratic volatility for straddle strategies and implied volatility term spread for strangle strategies respectively.

4 Robustness of options strategy momentum

In our earlier analysis, we employed several methods to assess the momentum patterns within the option strategies of straddle, strangle, and butterfly. The results consistently demonstrated the presence of a momentum effect in both straddle and strangle strategies, notably in the portfolio with a formation period spanning the past 24 months, excluding the most recent month, which yields the highest and most statistically significant returns. This section now extends our investigation to examine whether this momentum effect persists when constructing portfolios using alternative

approaches.

In earlier empirical analysis, we constructed a straddle using a put/call pair with the same strike that is closest to at-the-money, with the weights set to be delta-neutral; whereas a strangle consists of a weighted pair of out-of-the-money put/call options with strikes next to the one closest to at-the-money on both sides. These are both suitable strategies for an investor who expects a large move in the stock price in either direction. The strike of a straddle is typically selected to be the one closest to the underlying stock price,¹¹ but the selection of strangle strikes is more flexible. As the strike prices move further away from the underlying stock price, the costs of the strangle decrease, and consequently, the underlying stock price must make a greater move to yield a profit. Deep out-of-the-money options are also commonly associated with lower liquidity. Thus, in this section, we construct a strangle using further out-of-the-money put/call pairs with strikes next to the pair that we use in Section 3. We examine whether the momentum effects are robust to the selection of strikes in strangles.

Similar to our empirical analysis in Section 3, we begin with univariate sorts to assess the existence of momentum in the cross-section. Then, we examine the momentum effects after controlling for other option return predictors, from the perspectives of both portfolios and individual options.

Table 7 presents the results of univariate sorts in our robustness check. Consistent with the previous results in Table 1, we observe an evident monotonicity pattern in the tertile portfolio returns across almost all formation periods. In particular, for the skip-one-month formation periods over the past one to three years, the returns of the tertile portfolios consistently rise from the “Low” category to the “Medium” category and eventually reach their peak levels in the “High” category. As with prior results, the formation period of the past 2 to 24 months provides the strongest monthly return of 28.027% ($t=6.74$) with similar patterns emerging for other formation periods. For instance, the monthly average returns from formations (2, 12), (2, 36), and (13, 24) achieve 23.065% ($t = 4.68$), 25.511% ($t = 5.29$), and 15.911% ($t = 3.75$) respectively. These returns are all stronger than those of the original strangle strategies with higher levels of significance. Therefore, we argue that employing options with strikes further away from the at-the-money level leads to more significant tertile portfolio returns, primarily due to the increased occurrence of -100% returns when the options expire out-of-the-money.

We also examine momentum effects while controlling for other option return predictors. Table 8 presents the time-series regression results of strangle momentum returns using a linear pricing

¹¹Otherwise, either the call or put will be in-the-money, which can be costly and less liquid.

model comprising the five factors described in Section 3.2. Similar to the findings from univariate sorts, the strangle momentum for all formations, except for the reversal (1, 1), exhibit substantially stronger risk-adjusted monthly returns compared to the previous results presented in Table 3, with higher levels of statistical significance. In particular, the strangle momentum with the (2, 24) formation achieves the highest monthly return at 24.528% ($t=6.37$) among all periods. This result is consistent with the earlier findings presented in Table 3. The explanatory power of the idiosyncratic volatility (“IdiosynVol”) persists across fewer formation periods in (2, 24), (2, 36), as well as the long formation periods of (13, 36) and (25, 36). The smirk slope (“SmirkSlope”), on the other hand, shows explanatory power across slightly fewer formation periods than the results in Table 3. Other variables, including the volatility spread (“IV-HV”), the market capitalization of the underlying stock (“MktCap”), and the term spread of implied volatility (“TermSpread”) do not show any significant explanatory power. Generally, the stronger risk-adjusted returns and the significance observed across limited formation periods indicate a reduced capacity of the five factors to explain the momentum returns.

From the perspective of individual options, Table 9 presents the results from cross-sectional regressions of risk-adjusted individual option returns on past returns. The coefficients of past returns across nearly all formation periods exhibit notably higher levels of statistical significance when compared to the results presented in Table 5. This observation underscores the significant explanatory power of past returns in relation to individual strangle returns, even after accounting for the influence of the five predetermined predictors. Consequently, this finding provides robust evidence of the existence of momentum effects within the context of individual options, further reinforcing the notion that past returns play a significant role in shaping these returns.

Interestingly, we observe unusually large negative R^2 values. In addition to the increased frequency of -100% returns, we also find that strangles employing deep out-of-the-money options tend to generate more abnormally large positive returns. This occurs when a low initial premium turns into a substantial profit. The interplay of these factors results in distorted linear regression outcomes, as evidenced by the abnormal R^2 values. Thus, without imposing a linear structure, the results of conditional two-way sorts from Table 10 demonstrate the existence of momentum from an alternative perspective. After controlling for each corresponding factor, we observe monotonous tertile portfolio returns as well as greater and more significant high-minus-low momentum returns compared to the previous findings shown in Table 6.

Table 7: **Univariate sorts of strangle returns, robustness check.** This table reports the means and t -statistics for the monthly returns of the strangle portfolios in the robustness check. The underlying stocks are sorted into tertile portfolios each month based on the average returns during the formation period shown in the row header. The return is computed based on the equal-weighted strategy returns of the underlying stocks. The t -statistics in parentheses are corrected using Newey-West standard errors with three lags.

Formation period	Strangles			
	Low	Medium	High	High-Low
(1, 1)	-63.524 (-14.37)	-65.516 (-19.03)	-57.511 (-12.14)	6.013 (1.56)
(1, 2)	-67.370 (-18.69)	-67.374 (-16.95)	-53.064 (-10.77)	14.306 (3.57)
(1, 6)	-72.815 (-19.83)	-63.218 (-14.78)	-52.090 (-10.23)	20.725 (4.24)
(1, 12)	-76.089 (-19.63)	-62.812 (-14.82)	-50.050 (-9.32)	26.038 (4.94)
(2, 12)	-75.229 (-18.23)	-60.588 (-11.96)	-52.164 (-10.18)	23.065 (4.68)
(2, 24)	-77.848 (-26.53)	-59.321 (-11.69)	-49.821 (-9.25)	28.027 (6.74)
(2, 36)	-74.295 (-22.90)	-60.914 (-10.85)	-48.784 (-9.03)	25.511 (5.29)
(13, 24)	-71.737 (-17.65)	-59.626 (-12.10)	-55.826 (-11.40)	15.911 (3.75)
(13, 36)	-70.232 (-17.30)	-61.192 (-13.37)	-52.909 (-10.63)	17.323 (4.27)
(25, 36)	-65.749 (-14.76)	-59.424 (-10.55)	-58.395 (-11.92)	7.354 (1.72)

5 Conclusion

In this paper, we revisit the presence of cross-sectional momentum within the options market, specifically focusing on options associated with constituent firms of the S&P 100 Index. Our analysis centers on three commonly traded delta-neutral option strategies: straddles, strangles, and butterfly spreads.

We observe significant momentum returns in both straddle and strangle strategies. Firms whose options demonstrate robust performance over the preceding periods of 2 to 12 months, 2 to 24 months, and 2 to 36 months are more likely to experience heightened option returns in the subsequent month. Notably, among these, the momentum effect associated with a 2 to 24-month formation period yields the highest return with the most significant level when compared to other formation periods. This result is consistent for both straddle and strangle strategies. Moreover, it is noteworthy that the momentum returns obtained through both straddle and strangle strategies exhibit greater strength and statistical significance compared to those documented in Heston et al. (2023). This indicates that realizing returns from the options momentum effect can be accomplished with considerably smaller subsets of liquid options, particularly those associated with large-cap

Table 8: **Risk-adjusted strangle momentum returns, robustness check.** This table reports the results from time-series regressions of strangle momentum returns in the robustness check on a linear pricing model consisting of the five factors described in section 3.2. The momentum returns are computed based on a series of formation periods. Each row presents the regression results of a particular formation period given in the first column. The t-statistics in parentheses are corrected using Newey-West standard errors with three lags.

Formation period	Strangles						R^2
	Intercept	IV-HV	IdiosynVol	MktCap	TermSpread	SmirkSlope	
(1, 1)	4.676 (0.99)	-0.143 (-1.26)	0.194 (1.41)	0.108 (0.78)	-0.026 (-0.33)	-0.207 (-1.70)	0.089
(1, 2)	11.108 (2.60)	-0.069 (-0.61)	0.295 (2.22)	0.209 (1.46)	-0.086 (-0.72)	-0.355 (-2.35)	0.193
(1, 6)	15.416 (3.51)	0.321 (1.34)	0.069 (0.19)	-0.368 (-1.04)	0.034 (0.27)	-0.272 (-2.90)	0.204
(1, 12)	18.794 (4.00)	0.231 (1.28)	0.327 (1.19)	-0.112 (-0.47)	0.096 (0.55)	-0.254 (-2.91)	0.253
(2, 12)	19.038 (3.98)	0.231 (1.35)	0.394 (1.58)	-0.229 (-0.94)	0.149 (0.94)	-0.183 (-2.22)	0.285
(2, 24)	24.528 (6.37)	0.148 (1.18)	0.411 (3.56)	-0.039 (-0.31)	0.157 (1.38)	-0.085 (-0.79)	0.276
(2, 36)	23.961 (4.85)	0.048 (0.23)	0.484 (2.79)	-0.118 (-0.95)	0.065 (0.55)	0.110 (0.88)	0.179
(13, 24)	18.441 (3.92)	-0.046 (-0.38)	0.064 (0.50)	-0.081 (-0.65)	0.197 (2.26)	0.058 (0.59)	0.052
(13, 36)	18.592 (5.06)	-0.063 (-0.48)	0.330 (2.54)	0.031 (0.22)	0.141 (1.76)	0.232 (1.71)	0.144
(25, 36)	11.220 (2.56)	0.072 (0.72)	0.366 (3.17)	0.142 (1.12)	-0.066 (-0.80)	0.340 (3.56)	0.265

and blue-chip firms, as exemplified by the constituent companies within the S&P 100 Index. This stands in contrast to the broader approach of encompassing the entire spectrum of U.S. equities, as suggested by Heston et al. (2023).

These significant returns continue to be robust even after controlling for the five factors derived from conventional option risk characteristics. Of these five factors, only idiosyncratic volatility (referred to as “IdiosynVol”) and the smirk slope of implied volatility (referred to as “SmirkSlope”) exhibit noteworthy explanatory strength concerning the momentum returns associated with both straddle and strangle strategies, particularly during the formation periods of (2, 12) and (2, 24). Conversely, the remaining factors display limited explanatory capacity in relation to the momentum returns across nearly all formation periods. The significant explanatory power of past returns to the risk-adjusted returns of individual options strategies also continues to affirm the presence of momentum effects within straddle and strangle strategy returns.

We further document that both straddle and strangle strategies yield stronger returns, exhibiting higher levels of statistical significance when considered within the neutral context of each of the five factors, respectively. These returns, as observed across tertiles, reveal a distinct and clear

Table 9: **Cross-sectional regressions of risk-adjusted individual option returns, robustness check.** This table reports the results from cross-sectional regressions of risk-adjusted individual option returns in the robustness check on past returns, using the Fama-MacBeth procedure. The individual option returns are strategy returns for individual underlying stocks based on straddles, strangles, and butterfly spreads, respectively. The past returns are computed based on a series of formation periods. Each row presents the regression results of a particular formation period given in the first column. The t-statistics in parentheses are corrected using Newey-West standard errors with three lags.

Formation period	Strangles		
	Intercept	PastReturn	$\bar{R}^2$
(1, 1)	-67.566 (-16.07)	0.112 (3.66)	-6.934
(1, 2)	-73.323 (-14.77)	0.061 (1.48)	-16.879
(1, 6)	-61.407 (-10.41)	0.214 (4.10)	-19.649
(1, 12)	-61.717 (-12.58)	0.237 (5.81)	-35.225
(2, 12)	-62.574 (-12.99)	0.226 (5.81)	-8.121
(2, 24)	-55.998 (-10.05)	0.277 (5.68)	-25.490
(2, 36)	-53.050 (-8.60)	0.313 (5.64)	-33.054
(13, 24)	-66.489 (-13.32)	0.126 (3.03)	-16.208
(13, 36)	-62.491 (-11.23)	0.180 (3.87)	-24.970
(25, 36)	-66.973 (-13.07)	0.121 (3.13)	-35.573

Table 10: **Double sorts at formation period of lags 2 to 24, robustness check.** This table reports the results of conditional two-way sorts for the strangle returns in the robustness check. For each month, we first sort stocks into tertiles based on a control variable shown in the row header. After that, within each tertile, we sort the stocks, again into tertiles, based on the average strangle returns during the formation period of the past 2-to-24 months and construct an equal-weighted portfolio. We then take the average of the past-return-based portfolio returns over the control variable tertiles. Each row presents the means and t-statistics, in parentheses, of the past-return-sorted tertile portfolio returns after controlling for the corresponding control variable. The t-statistics are corrected using Newey-West standard errors with three lags.

Control	Strangles			
	Low	Medium	High	High-Low
IV-HV	-71.618 (-19.87)	-67.282 (-14.85)	-51.032 (-9.23)	42.968 (7.66)
IdiosynVol	-72.778 (-21.82)	-65.470 (-16.02)	-54.998 (-10.98)	40.455 (6.50)
MktCap	-74.185 (-18.53)	-64.078 (-14.34)	-52.382 (-9.99)	39.858 (5.85)
TermSpread	-76.475 (-19.61)	-66.558 (-12.75)	-54.573 (-10.09)	39.931 (5.62)
SmirkSlope	-69.570 (-19.83)	-67.217 (-15.15)	-51.389 (-10.23)	37.417 (5.68)

monotonic increasing trend, progressing from the “Low” category to the “Medium” category and ultimately reaching their highest levels in the “High” category. Notably, within straddle strategies,

the long positions yield significantly higher returns compared to the short positions, whereas in strangle strategies, the short positions exhibit dominance.

Finally, it is noteworthy that we observe a short-term reversal phenomenon in straddles, which does not attain statistical significance in the case of strangles. We also do not find substantial evidence supporting momentum effects, although there is a marginally significant indication of reversal in butterfly spread returns.

Although robust momentum returns are observed in straddle and strangle returns, the fundamental cause of these empirical findings remains uncertain. If these returns occur as compensation for an unknown risk factor, it is up to future research to develop a cross-sectional options return model that subsumes momentum profitability. On the other hand, if it turns out to be abnormal, the underlying cause of this behaviour is left for future studies to address. Moreover, we document distinctions in the momentum patterns between straddles and strangles, prompting further inquiry into the reasons behind these disparities.

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